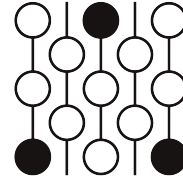


Review of Submission 3D

First Proof Second Batch

Reviewer 1

June 4–8, 2026



1 Review

The submission proves that the inequality holds for $p = 1/k$, $k \in \mathbb{Z}_{\geq 0}$. The argument is the same as the argument of N. Alon, Y. Emek, M. Feldman, and M. Tennenholtz in their paper *Adversarial Leakage in Games*.

Correctness. The submission correctly proves the inequality for $p = 1/k$ for $k \in \mathbb{Z}_{>0}$ using a well-known proof without proper citation. The submission does not consider other values of p .

Novelty. The presented argument appears in other work and is not cited.

2 Recommendation

This submission should be **rejected**. It repeats a known argument without citation, and does not attempt to answer the question beyond this point.

3 Submission with inline comments

Problem Statement

Let $w_1, \dots, w_m$ be non-negative real numbers such that $\sum_{i=1}^m w_i = 1$, and let $v_1, \dots, v_m$ be i.i.d. Bernoulli(p) random variables. For which values of probability p is it true that

$$\Pr \left[\sum_{i=1}^m w_i v_i \geq p \right] \geq p?$$

Partial Solution

Note: The following represents a partial solution to the problem. As noted at the end of the proof, sufficiency is fully established, but there remains a structural gap in proving the necessity for non-reciprocal values of p .

This argument appears in work of N. Alon, Y. Emek, M. Feldman, and M. Tennenholtz.

Theorem 1. *Let $w_1, \dots, w_m$ be non-negative real numbers such that $\sum_{i=1}^m w_i = 1$, and let $v_1, \dots, v_m$ be i.i.d. Bernoulli(p) random variables. The inequality*

$$\Pr \left[\sum_{i=1}^m w_i v_i \geq p \right] \geq p$$

holds for all valid weight distributions if and only if $p = 1/k$ for some integer $k \geq 1$ (or $p = 0$).

Proof. For $p = 0$, the condition reduces to $\Pr[0 \geq 0] \geq 0$, which evaluates to $1 \geq 0$ and is trivially true.

Assume $p = 1/k$ for some integer $k \geq 1$. We construct k coupled sequences of random variables $v_i^{(j)}$ for indices $i \in \{1, \dots, m\}$ and $j \in \{1, \dots, k\}$ as follows: For each i , select an index J_i uniformly at random from the set $\{1, \dots, k\}$, such that the choices $J_1, \dots, J_m$ are mutually independent. Set $v_i^{(J_i)} = 1$, and set $v_i^{(j)} = 0$ for all $j \neq J_i$.

By this construction, for any fixed sequence index j , the variables $v_1^{(j)}, \dots, v_m^{(j)}$ are independent Bernoulli($1/k$) random variables. Define $X^{(j)} = \sum_{i=1}^m w_i v_i^{(j)}$. For each j , the random variable $X^{(j)}$ has the exact same distribution as the original sum $\sum_{i=1}^m w_i v_i$.

Summing across all k sequences yields:

$$\sum_{j=1}^k X^{(j)} = \sum_{j=1}^k \sum_{i=1}^m w_i v_i^{(j)} = \sum_{i=1}^m w_i \sum_{j=1}^k v_i^{(j)}$$

Since exactly one $v_i^{(j)} = 1$ for each i (specifically when $j = J_i$), the inner sum $\sum_{j=1}^k v_i^{(j)}$ equals 1 deterministically. Thus,

$$\sum_{j=1}^k X^{(j)} = \sum_{i=1}^m w_i (1) = 1$$

Because the k identically distributed random variables $X^{(j)}$ sum to exactly 1, at least one of them must evaluate to $1/k$ or greater. Let A_j be the event that $X^{(j)} \geq 1/k$. We have $\Pr \left[\bigcup_{j=1}^k A_j \right] = 1$. By the union bound:

$$1 = \Pr \left[\bigcup_{j=1}^k A_j \right] \leq \sum_{j=1}^k \Pr[A_j] = k \Pr \left[X^{(1)} \geq \frac{1}{k} \right]$$

Dividing by k gives:

$$\Pr \left[\sum_{i=1}^m w_i v_i \geq \frac{1}{k} \right] \geq \frac{1}{k}$$

This establishes sufficiency.

There are more values of p which satisfy the inequality universally.

For values of $p \notin \{0\} \cup \{1/k \mid k \in \mathbb{Z}^+\}$, the inequality does not universally hold.

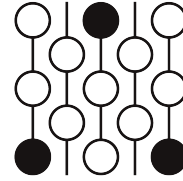
[Gap: The explicit construction of pathological weight distributions providing counterexamples for all non-reciprocal values of p is structurally omitted.] $\square$

Review of Submission 3D

First Proof Second Batch

Reviewer 2

June 4–8, 2026



1 Review

Correctness. I recommend **rejecting this submission**. The proof of Theorem 1 is correct. However, it is essentially a direct replication of the simple coupling argument from the paper by Alon, Emek, Feldman, and Tennenholtz (see Remark on page 7 of <https://web.math.princeton.edu/~nalon/PDFS/leakage7.pdf>). The argument is not developed any further to yield meaningful progress on the problem. In particular, it does not rule out any values of p , and the submission ultimately arrives at an incorrect conjecture for the answer.

Novelty. None. The main argument is the same as the coupling argument in the paper cited above and does not contain any new ideas or insights.

2 Recommendation

3 Submission with inline comments

Problem Statement

Let $w_1, \dots, w_m$ be non-negative real numbers such that $\sum_{i=1}^m w_i = 1$, and let $v_1, \dots, v_m$ be i.i.d. Bernoulli(p) random variables. For which values of probability p is it true that

$$\Pr \left[\sum_{i=1}^m w_i v_i \geq p \right] \geq p?$$

Partial Solution

Note: The following represents a partial solution to the problem. As noted at the end of the proof, sufficiency is fully established, but there remains a structural gap in proving the necessity for non-reciprocal values of p .

Theorem 1. Let $w_1, \dots, w_m$ be non-negative real numbers such that $\sum_{i=1}^m w_i = 1$, and let $v_1, \dots, v_m$ be i.i.d. Bernoulli(p) random variables. The inequality

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holds for all valid weight distributions if and only if $p = 1/k$ for some integer $k \geq 1$ (or $p = 0$).

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This establishes sufficiency.

For values of $p \notin \{0\} \cup \{1/k \mid k \in \mathbb{Z}^+\}$, the inequality does not universally hold.

[Gap: The explicit construction of pathological weight distributions providing counterexamples for all non-reciprocal values of p is structurally omitted.] $\square$